

AIDataCenter

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Specification and bootstrap stack for a **64× NVIDIA B200**

research AI inference cluster.

1 Contents

Path	Description
SPEC.md	Full facility / fabric / storage / K8s design
docs/	Build guide, network (ZTP/RoCE), overlay (EVPN/VXLAN), serving (Kimi K2 + API), bootstrap, NetBox SoT, cabling guide
bootstrap/	Seed host, Metal3/CAPI, Flux platform, NetBox
demo/	Interactive live bring-up simulator (browser)
docs/training-guide/	<i>The AI Datacenter Network</i> — 12-chapter training guide on RDMA fabrics, lossless Ethernet, and GPU collective traffic

2 Highlights

- **4 racks:** GPU-1, GPU-2, BOOT, STOR
- **Fabric:** Arista 7060DX5 rail-optimized 8×400G/node (RoCEv2)
- **OOB:** Arista 7010TX-48
- **Platform:** Kubernetes + NVIDIA GPU Operator + KServe + Ray
- **Serving:** Kimi K2 Thinking (TP=8 × replicas) behind Envoy gateway + inference scheduler
(docs/serving.md)
- **Overlay:** EVPN/VXLAN — vrf storage + vrf edge for API ingress (docs/overlay.md)
- **Source of truth:** NetBox for inventory, IPAM, and cabling
- **GitOps:** Flux-managed platform apps (Harbor, Vault, observability, Redpanda, ...)

3 Quick start

```
# Interactive demo (simulated bootstrap → switches → GPU cluster)

python3 -m http.server 8765 --directory demo

# open http://127.0.0.1:8765


# Design-time inventory + cabling (no hardware required)

bash bootstrap/scripts/netbox-sync.sh --offline


# Physical build: docs/build-guide.md → then docs/bootstrap.md + bootstrap/README.md
```

4 Documentation (PDF)

Every Markdown doc has a Typst twin + compiled PDF next to it
(e.g. docs/build-guide.typ → docs/build-guide.pdf, README.pdf).

Regenerate all after editing any .md source:

```
pip install pypandoc-binary # plus typst CLI on PATH
python3 docs/md2pdf.py      # or: python3 docs/md2pdf.py docs/build-guide.md
```

Styling is shared via docs/template.typ; generated .typ
files are overwritten on each run — edit the .md or the template,
not the generated output.
docs/training-guide/ is the exception: it is authored directly
in Typst (no Markdown source, own
lib.typ styling) and is not part
of the md2pdf.py set. Build it with:

```
typst compile --root . docs/training-guide/main.typ \
docs/training-guide/AI-Datacenter-Network-Training-Guide.pdf
```

5 License

Add a license before production use. Content is provided as-is for design and operations planning.